THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 60°	Complementary angles add up to 90°.
2	**(a)** 130°	Supplementary angles add up to 180°.
3	**(b)** 40°	Vertically opposite angles are equal.
4	**(c)** 120°	A linear pair adds up to 180°.
5	**(c)** parallel lines	Recall the definition.
6	**(d)** 55°	Corresponding angles are equal when the lines are parallel.
7	**(b)** 48°	Alternate interior angles are equal.
8	**(a)** 25°	Complementary angles add up to 90°.
9	**(b)** 65°	Supplementary angles add up to 180°.
10	**(d)** 75°	Vertically opposite angles are equal.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Complement $= 90^\circ - 35^\circ = 55^\circ$ (1); Supplement $= 180^\circ - 110^\circ = 70^\circ$ (1).	1+1
12	Vertically opposite $= 65^\circ$ (1); Other two $= 180^\circ - 65^\circ = 115^\circ$ each (1).	1+1
13	Corresponding angle $= 70^\circ$ (1); Co-interior angles are supplementary: 110° (1).	1+1
14	Complement $= 90^\circ - 62^\circ = 28^\circ$ (1); Supplement $= 180^\circ - 75^\circ = 105^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum $= 180^\circ$, total parts 3 (1); One part $= 60^\circ$ (1); Angles 60° and 120° (1).	1+1+1
16	$x + 110 = 180$ (1); $x = 70$ (1); Angle is acute (1).	1+1+1
17	Angle 2 $= 52^\circ$ (alternate interior) (1); Co-interior $= 180^\circ - 52^\circ = 128^\circ$ (1); Reason: parallel lines (1).	1+1+1
	OR: Sum $= 180^\circ$, total parts 6 (1); One part $= 30^\circ$ (1); Angles 30° and 150° (1).	1+1+1
18	$x + 135 = 180$ (1); $x = 45$ (1); Angle is acute (1).	1+1+1
	OR: Angle 2 $= 66^\circ$ (alternate interior) (1); Co-interior $= 180^\circ - 66^\circ = 114^\circ$ (1); Reason: parallel lines (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$x = 145$ (1); Vertically opposite pairs: $35^\circ, 35^\circ$ and $145^\circ, 145^\circ$ (1); Sum check: 360° (1); Four angles around a point make 360° (2).	1+1+1+2
	OR: Angles equal to 40°: vertically opposite, corresponding, alternate (4 in all) (1); Angles equal to 140°: the other 4 (1); Linear pair: $40 + 140 = 180$ (1); State reasons for each (2).	1+1+1+2
20	$x = 130$ (1); Vertically opposite pairs: $50^\circ, 50^\circ$ and $130^\circ, 130^\circ$ (1); Sum check: 360° (1); Four angles around a point make 360° (2).	1+1+1+2
	OR: Angles equal to 55°: vertically opposite, corresponding, alternate (4 in all) (1); Angles equal to 125°: the other 4 (1); Linear pair: $55 + 125 = 180$ (1); State reasons for each (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 80°	A linear pair adds up to 180°.
2	**(b)** transversal	Recall the definition.
3	**(a)** 82°	Corresponding angles are equal when the lines are parallel.
4	**(c)** 73°	Alternate interior angles are equal.
5	**(b)** 42°	Complementary angles add up to 90°.
6	**(a)** 102°	Supplementary angles add up to 180°.
7	**(c)** 110°	Vertically opposite angles are equal.
8	**(c)** 55°	A linear pair adds up to 180°.
9	**(a)** complementary	Recall the definition.
10	**(c)** 118°	Corresponding angles are equal when the lines are parallel.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Vertically opposite $= 110^\circ$ (1); Other two $= 180^\circ - 110^\circ = 70^\circ$ each (1).	1+1
12	Corresponding angle $= 105^\circ$ (1); Co-interior angles are supplementary: 75° (1).	1+1
13	Complement $= 90^\circ - 18^\circ = 72^\circ$ (1); Supplement $= 180^\circ - 140^\circ = 40^\circ$ (1).	1+1
14	Vertically opposite $= 40^\circ$ (1); Other two $= 180^\circ - 40^\circ = 140^\circ$ each (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum $= 180^\circ$, total parts 5 (1); One part $= 36^\circ$ (1); Angles 72° and 108° (1).	1+1+1
16	$x + 150 = 180$ (1); $x = 30$ (1); Angle is acute (1).	1+1+1
17	Angle 2 $= 74^\circ$ (alternate interior) (1); Co-interior $= 180^\circ - 74^\circ = 106^\circ$ (1); Reason: parallel lines (1).	1+1+1
	OR: Sum $= 180^\circ$, total parts 9 (1); One part $= 20^\circ$ (1); Angles 80° and 100° (1).	1+1+1
18	$x + 95 = 180$ (1); $x = 85$ (1); Angle is acute (1).	1+1+1
	OR: Angle 2 $= 88^\circ$ (alternate interior) (1); Co-interior $= 180^\circ - 88^\circ = 92^\circ$ (1); Reason: parallel lines (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$x = 115$ (1); Vertically opposite pairs: $65^\circ, 65^\circ$ and $115^\circ, 115^\circ$ (1); Sum check: 360° (1); Four angles around a point make 360° (2).	1+1+1+2
	OR: Angles equal to 70°: vertically opposite, corresponding, alternate (4 in all) (1); Angles equal to 110°: the other 4 (1); Linear pair: $70 + 110 = 180$ (1); State reasons for each (2).	1+1+1+2
20	$x = 100$ (1); Vertically opposite pairs: $80^\circ, 80^\circ$ and $100^\circ, 100^\circ$ (1); Sum check: 360° (1); Four angles around a point make 360° (2).	1+1+1+2
	OR: Angles equal to 85°: vertically opposite, corresponding, alternate (4 in all) (1); Angles equal to 95°: the other 4 (1); Linear pair: $85 + 95 = 180$ (1); State reasons for each (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 99°	Alternate interior angles are equal.
2	**(a)** 18°	Complementary angles add up to 90°.
3	**(c)** 40°	Supplementary angles add up to 180°.
4	**(d)** 135°	Vertically opposite angles are equal.
5	**(a)** 145°	A linear pair adds up to 180°.
6	**(d)** perpendicular	Recall the definition.
7	**(a)** 67°	Corresponding angles are equal when the lines are parallel.
8	**(d)** 125°	Alternate interior angles are equal.
9	**(a)** 75°	Complementary angles add up to 90°.
10	**(b)** 85°	Supplementary angles add up to 180°.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Corresponding angle $= 58^\circ$ (1); Co-interior angles are supplementary: 122° (1).	1+1
12	Complement $= 90^\circ - 47^\circ = 43^\circ$ (1); Supplement $= 180^\circ - 98^\circ = 82^\circ$ (1).	1+1
13	Vertically opposite $= 125^\circ$ (1); Other two $= 180^\circ - 125^\circ = 55^\circ$ each (1).	1+1
14	Corresponding angle $= 132^\circ$ (1); Co-interior angles are supplementary: 48° (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum $= 180^\circ$, total parts 4 (1); One part $= 45^\circ$ (1); Angles 45° and 135° (1).	1+1+1
16	$x + 70 = 180$ (1); $x = 110$ (1); Angle is obtuse (1).	1+1+1
17	Angle 2 $= 103^\circ$ (alternate interior) (1); Co-interior $= 180^\circ - 103^\circ = 77^\circ$ (1); Reason: parallel lines (1).	1+1+1
	OR: Sum $= 180^\circ$, total parts 9 (1); One part $= 20^\circ$ (1); Angles 40° and 140° (1).	1+1+1
18	$x + 160 = 180$ (1); $x = 20$ (1); Angle is acute (1).	1+1+1
	OR: Angle 2 $= 117^\circ$ (alternate interior) (1); Co-interior $= 180^\circ - 117^\circ = 63^\circ$ (1); Reason: parallel lines (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$x = 65$ (1); Vertically opposite pairs: $115^\circ, 115^\circ$ and $65^\circ, 65^\circ$ (1); Sum check: 360° (1); Four angles around a point make 360° (2).	1+1+1+2
	OR: Angles equal to 100°: vertically opposite, corresponding, alternate (4 in all) (1); Angles equal to 80°: the other 4 (1); Linear pair: $100 + 80 = 180$ (1); State reasons for each (2).	1+1+1+2
20	$x = 50$ (1); Vertically opposite pairs: $130^\circ, 130^\circ$ and $50^\circ, 50^\circ$ (1); Sum check: 360° (1); Four angles around a point make 360° (2).	1+1+1+2
	OR: Angles equal to 125°: vertically opposite, corresponding, alternate (4 in all) (1); Angles equal to 55°: the other 4 (1); Linear pair: $125 + 55 = 180$ (1); State reasons for each (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.